

# LUYOLO SKOMA

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Software Engineer | AI Integration | Full-Stack | Python | JavaScript | React / React Native  
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## PROFESSIONAL PROFILE

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Advanced Diploma graduate in ICT: Applications Development (CPUT, NQF Level 7) with production-grade full-stack engineering experience across web and mobile platforms serving real business clients. Appointed Team Lead at Plum Systems, owning delivery across a cross-functional team of 15 developers on two simultaneous production projects demonstrating accountability, technical leadership, and the ability to perform under real delivery pressure. Architected and shipped AI-powered features using the OpenAI LLM API in live production environments, translating cutting-edge technology into scalable, user-facing solutions. Brings deep expertise in object-oriented programming, clean architecture, REST API design, SQL databases, and cloud services (Firebase / GCP). Consistently acquires and applies emerging technologies rapidly not in isolation, but in production, at scale, and under real business constraints.

## PROFESSIONAL EXPERIENCE

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**Junior Software Developer** | Plum Systems | Bellville, Cape Town June 2024 – March 2026  
*Post-WIL permanent role: joined full-time after completing Advanced Diploma requirements, while finalising qualification. Zero months post-degree experience - graduated April 2026.*

- ▶ Took on expanded technical ownership in a production environment serving real business clients, building, and maintaining full-stack features across commercial and industrial property platforms available on both web and mobile.
- ▶ Integrated OpenAI LLM API into a live production web application, enabling AI-powered workflow automation for end users, gaining direct, hands-on experience translating AI/ML capabilities into scalable product features.
- ▶ Designed and built scalable REST APIs using Firebase Cloud Functions to support both frontend web and React Native mobile applications, improving platform data handling and reducing client-side load.
- ▶ Contributed to frontend and backend architecture improvements across production systems, improving application performance, responsiveness, and maintainability for business clients.
- ▶ Maintained and enhanced live systems used by paying business clients, ensuring high availability and responsiveness to production issues.
- ▶ Tech: JavaScript, React, React Native, Node.js, Firebase (Auth, Firestore, Functions), SQL(PostgreSQL), Git, GitHub, Google Cloud Platform, OpenAI API
- ▶ Assisted with system data migration and backend integration; contributed to GitHub collaboration workflows including pull requests, code reviews, and CI/CD pipeline deployment.
- ▶ Improved application performance, responsiveness, and data handling across both platforms; participated in ongoing production support and feature delivery.

## PRODUCTION PROJECTS

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**API Property** | Plum Systems: Production Mobile & Web Application Jun 2024 – Mar 2026

- ▶ Commercial and industrial property management platform available on web and mobile, serving real business clients. Contributed to full-stack development, REST API design, and ongoing production maintenance.
- ▶ Maintained and enhanced live, high-availability platforms serving paying business clients, ensuring system uptime and rapid response to production issues.

**Stack:** React, React Native, Node.js, Firebase, SQL, GCP

### **Officeplace** | *Plum Systems: Production Mobile & Web Application*

Jun 2024 – Mar 2026

- ▶ Officially appointed Team Lead - coordinated delivery across a 15-person cross-functional intern team; contributed to feature development, backend integration, data migration, and scalable service maintenance.
- ▶ Redesigned UI components in collaboration with cross-functional teams to improve usability, responsiveness, and end-user satisfaction across production platforms.
- ▶ Maintained and evolved live production platforms over a sustained 12+ month period, managing feature enhancements, performance improvements, and production issue resolution for paying business clients demonstrating long-term solution ownership beyond initial delivery.

**Stack:** React, React Native, Node.js, Firebase, SQL, GCP

## **PERSONAL PROJECTS**

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### **MetroLink Cape Town** | *Personal Project: Transport Intelligence Platform*

- ▶ Designed and developed a full-stack transport intelligence platform integrating Metrorail and MyCiTi services using .NET 9, ASP.NET Core Web API, React Native (Expo), PostgreSQL, and SignalR within an Agile 2-person team.
- ▶ Architected and implemented a RESTful API using clean architecture (Controllers, Services, Repositories) with Entity Framework Core (EF Core) and Npgsql, ensuring scalability, maintainability, and separation of concerns.
- ▶ Built a custom delay prediction and reliability scoring algorithm leveraging historical data, real-time inputs, and weighted calculations to generate route performance metrics and optimize user decisionmaking.
- ▶ Engineered a comprehensive security layer including JWT authentication with expiry validation, role-based access control (RBAC), BCrypt password hashing, CORS policy enforcement, Bearer token validation, and secure client-side storage mitigating common attack vectors and ensuring data confidentiality across all API endpoints.
- ▶ Applied security best practices including CORS configuration, Bearer token enforcement, secure client-side storage (AsyncStorage), and API endpoint protection.
- ▶ Created comprehensive technical documentation including API documentation, database ER diagrams, UML diagrams (sequence, activity), and system architecture diagrams, supporting stakeholder communication and system design validation.
- ▶ Produced comprehensive technical documentation including API references, database ER diagrams, UML sequence and activity diagrams, and system architecture diagrams to support maintainability and stakeholder communication
- ▶ Managed CI/CD pipelines and cloud deployment using GitLab, Railway, and PostgreSQL cloud hosting, including environment variable management, production configuration, and DATABASE\_URL parsing.

**Stack:** .NET 9 Web API, React Native (Expo), PostgreSQL 18, EF Core, SignalR, JWT Auth, Zustand, Axios, Railway, GitLab CI/CD, C#, JavaScript

### **Wonga Authentication System** | *Personal Project*

- ▶ Full-stack authentication system with clean service separation, JWT-based auth, BCrypt password hashing, and integration tests. Containerized with Docker Compose for streamlined local development and deployment.

**Stack:** React, .NET 9 Web API, PostgreSQL, Docker Compose

**Bruce Carter Store** | *Personal Project*

► Fully functional, responsive e-commerce web application demonstrating hands-on experience with frontend development, state management, authentication, and real-time database integration. **Stack:** React.js, Firebase, JavaScript

**Legends Barber** | *Personal Project*

- Mobile application connecting users to barber services, showcasing full-stack mobile expertise across navigation, state management, and Firebase integration.

**Stack:** React Native (Expo), Firebase

**TECHNICAL SKILLS**

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|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages<br>AI / ML       | JavaScript (ES6+), Python, Java, C#, SQL (PostgreSQL), HTML, CSS<br>OpenAI LLM API integration, AI-powered workflow automation, AI tooling in production development |
| Frameworks                 | React, React Native, Node.js, .NET 9 Web API, Firebase (Auth, Firestore, Cloud Functions)                                                                            |
| Databases                  | PostgreSQL, Firebase Firestore: schema design, EF Core migrations, querying, cloud data handling                                                                     |
| Cloud / DevOps             | GCP, Firebase Hosting, Railway, GitHub Actions, GitLab CI/CD, Docker Compose                                                                                         |
| Dev Tools                  | Git, GitHub, VS Code, Visual Studio, PyCharm, IntelliJ, Google Cloud Console                                                                                         |
| Multi-Tier<br>Architecture | Architected and implemented a scalable multi-tier system using a clean layered architecture (Controllers, Services, Repositories)                                    |
| OOP & CS                   | OOP (Java, C#, JavaScript), data structures, REST API design, scalable architecture, SignalR real-time                                                               |
| Platforms                  | Web Development, Mobile Development (iOS & Android via React Native)                                                                                                 |
| Security                   | JWT authentication, role-based authorisation, BCrypt hashing, CORS policy, secure token storage                                                                      |
| Algorithms                 | Delay reliability scoring, weighted historical analysis, peak-hour modelling, departure window prediction                                                            |
| Documentation              | API references, UML diagrams, database schemas, component guides, stakeholder presentations, Issue Testing Documentation                                             |

**EDUCATION**

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**Advanced Diploma in ICT: Applications Development** | *Cape Peninsula University of Technology (CPUT)* Jan 2025 – Jan 2026

- ▶ NQF Level 7 - formally equivalent to a bachelor's degree under South Africa's Higher Education Qualifications Sub-Framework (HEQSF). Graduated January 2026.
- ▶ Specialisation: Applications (Software) Development - full-stack engineering, software architecture, OOP, databases, and agile project delivery.

**Diploma in Information Technology** | *Cape Peninsula University of Technology (CPUT)* Jan 2022 – Jan 2025

- ▶ Foundation in software engineering, information systems, web and mobile development, databases, and networking.
- ▶ Completed Work Integrated Learning (WIL) placement at Plum Systems as final-year requirement.

**Higher Certificate in ICT** | *Cape Peninsula University of Technology (CPUT)*

2021

## CERTIFICATIONS

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- ▶ Google IT Support Professional Certificate - Coursera (2021)
  - ▶ Higher Certificate in ICT - CPUT (2021)

## REFERENCES

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**Anele Ngabeni** - Mentor / Team Lead, Plum Systems | 067 876 2327 *Additional references available upon request.*